# How to Be a Good AI Researcher

**Tao LIN**

September 23, 2025

WESTLAKE UNIVERSITY | SCHOOL OF ENGINEERING

# Acknowledgement

- The 7 Cs of Communication, World of Work Project
- Awesome Tips, JiaBin Huang
- 12 resolutions for grad students, Matt Might
- Tips for work-life balance, Matt Might
- Productivity tips for academics, Matt Might
- How I read research papers, Aaditya Ramdas
- GAMES003

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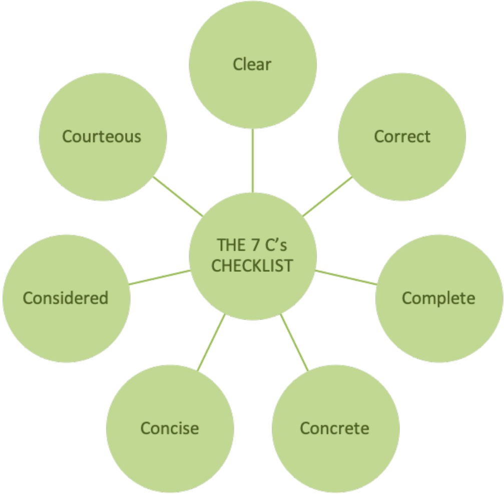
In the previous three lectures...

# We learn how to communicate: the 7 **C**'s of communication




Please revisit our previous lecture when necessary.

# We learn the principles of presentation

**Great talks require effort & time!**

# We learn the first course on “how to do research”

- The Illustrated Guide to a Ph.D.
- 10 Easy Ways to Fail a Ph.D.
- How to Make Steady Progress?
- How to Keep Track With the Literature?
- How to Read Papers?

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But why do we need to answer these questions?

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  - “I am citing this paper, so I should read it fully.”

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  I never read a paper from start to end on my first opening (or ever)

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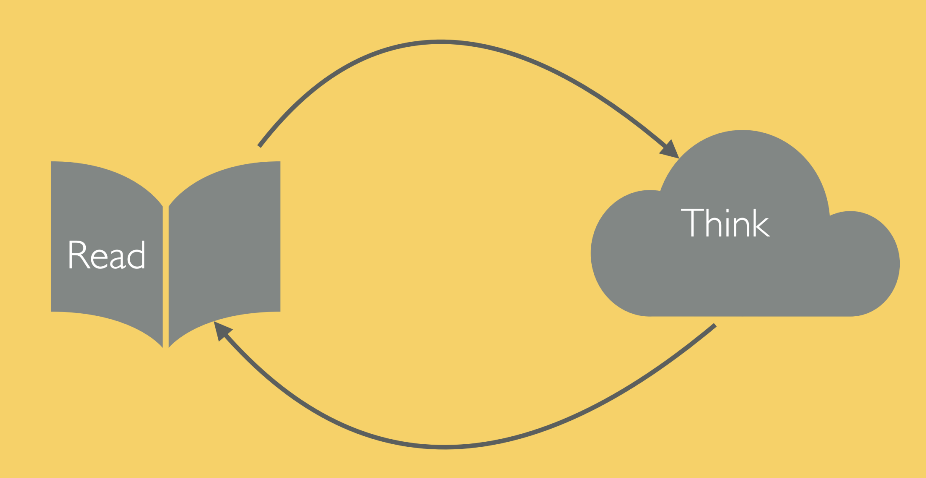
- Sometimes the reading needs to split across days

# The principle of iterative refinement



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**Sources**: abstract/intro, problem definition, main theorem, discussion.

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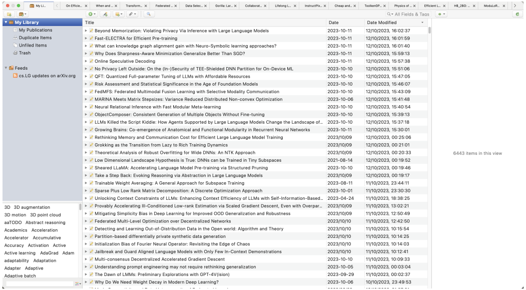
**Sources:** appendices, proof details, corollaries, remarks, related work

# How I organize my reading

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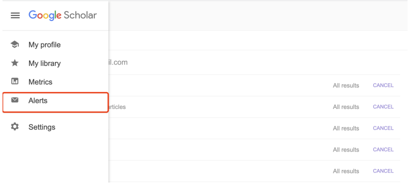
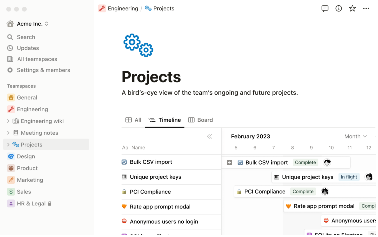



Google Scholar Alerts




Notion

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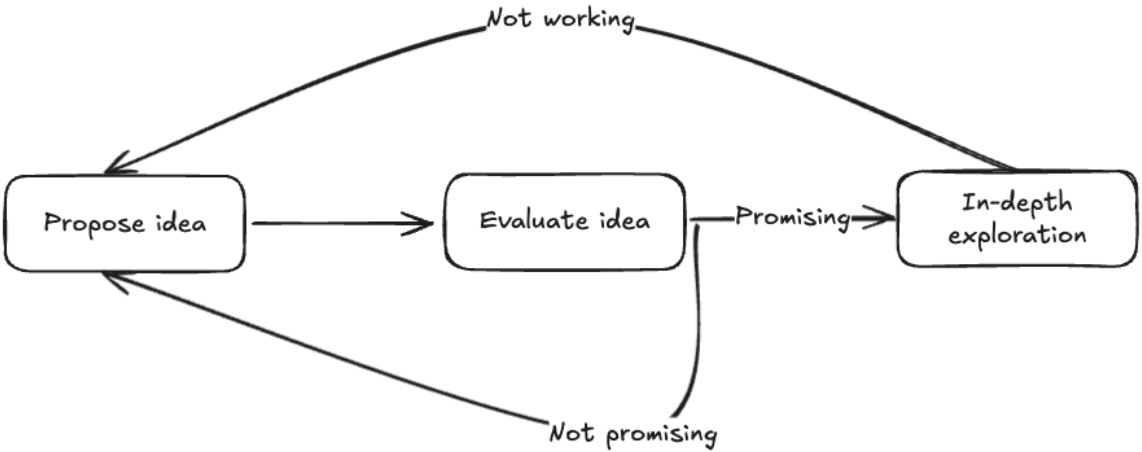
# A general loop



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  - Take a general approach and tailor it to your SPECIFIC problem.
  - Leverage all the domain knowledge (i.e., make more assumptions) to improve the method.

# Combine two ideas/problems

"To steal ideas from one person is plagiarism. To steal from many is research." — Wilson Mizner

# Add an adjective

Given an existing idea X, add an adjective to make it

- slow $\rightarrow$ fast
- batch $\rightarrow$ online
- sensitive $\rightarrow$ robust
- centralized $\rightarrow$ distributed
- single-step $\rightarrow$ progressive
- single-level $\rightarrow$ hierarchical
- fixed $\rightarrow$ adaptive, sth-aware
- data-hungry $\rightarrow$ data-efficient

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This is where your work can fill the gap.

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5. Critically consider your research taste, and the community taste around you. Your taste is likely very influenced by your research cluster (your collaborators, advisor, etc)

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Once we have the idea, we need to test them!


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Some tips on why, what, and how to do experiments.

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✓ Do an experiment to test a hypothesis.

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- break them down into baby steps
- design experiments that best answer those questions

# How?

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There are no universal answers, but here are some principles I found helpful.

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- Ask for feedback

  Your mentors/advisors are there to help you succeed.

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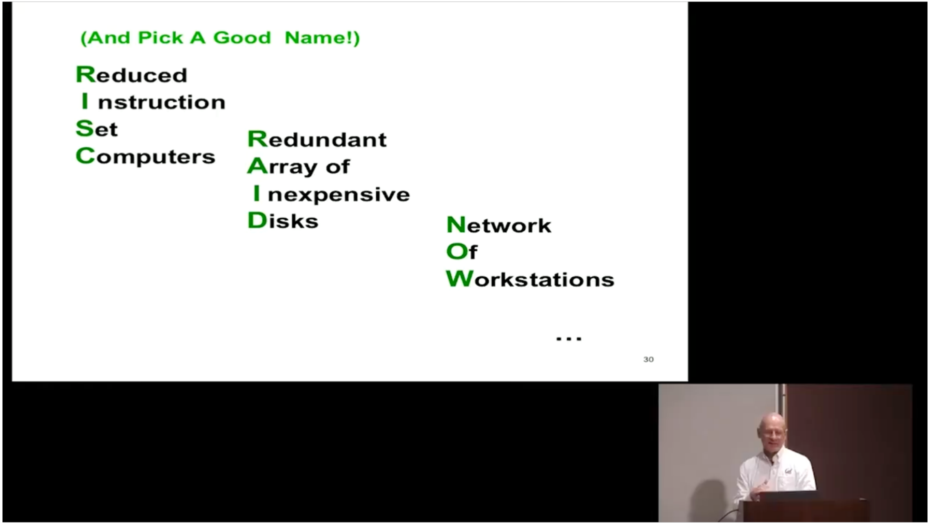
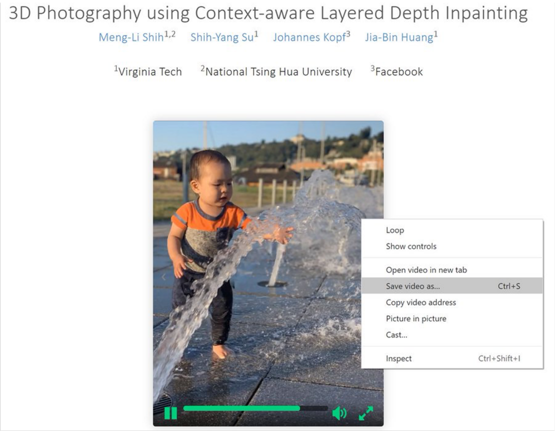
# Pick a good name



# Make all your results available




# Lower the barrier for others to follow

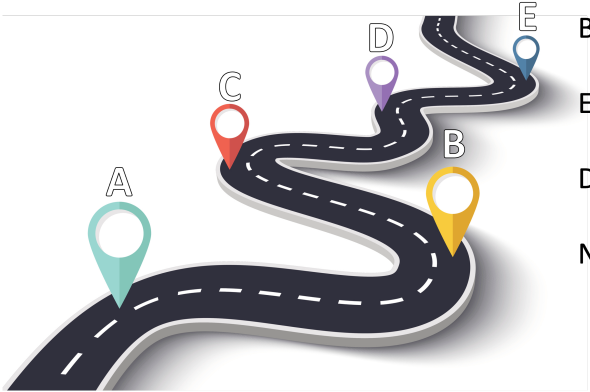
GitHub

Google colab

Hugging Face

PyTorch hub

# Make others' life easier



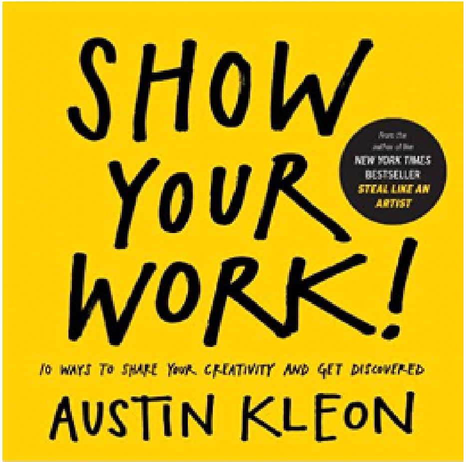

Baseline implementation?

Evaluation script?

Data preprocessing?

New dataset?

# Show your work!




Your website, Zhihu, Weibo, bilibili, YouTube, Twitter, etc

This week we will talk more on ***productivity and career*.**

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Map out what the next twelve months will look like.

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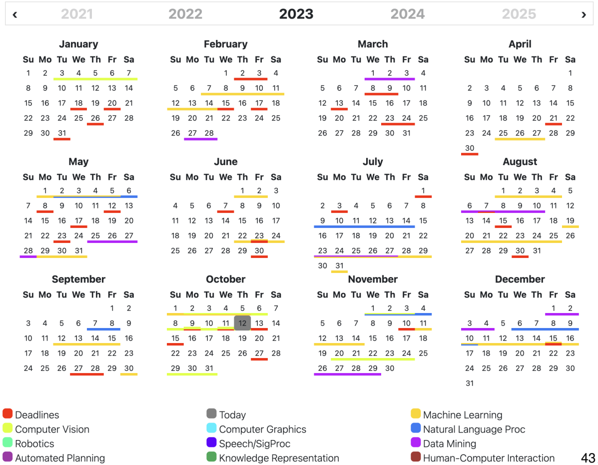

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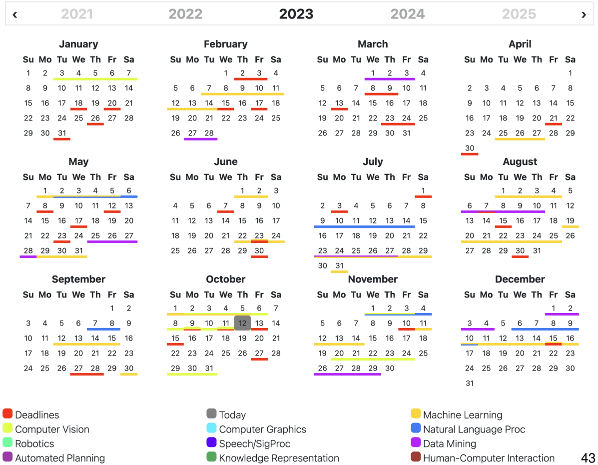

# Map out the year

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- Put major deadlines on your calendar.
- Decide program milestones

# Improve productivity

We will discuss it later!

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**Tips:**

- The “rule of 3”:

  When a third person recommends you try something, you must try it.

- The “15 minute rule”:

  Give something the benefit of the doubt for 15 minutes.
  If you don’t want to continue after 15 minutes, drop it.

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For example,

- Do you have an automatic research workflow?
- Is there software or hardware that could accelerate your workflow?
- Have you optimized your configuration files?
- Is it time to set up your LaTeX macros?

# Stay healthy

Mind and body are connected: a healthy body supports a creative mind.

[1] It's OK to get depressed. It's not OK to do nothing about it.

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# Stay healthy

Mind and body are connected: a healthy body supports a creative mind.

- Evaluate your diet and exercise habits.
- Learn to do something with your body.
- Watch out for mental health[1].

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# Update your CV and web site

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- Try your best to improve your profile / product.

# Keep your eye on the job market

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- If your school is interviewing, attend hiring talks.

# Network

Your future success will depend in part on networking.

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For example,

- You will need letter-writers.
- You will need them to pull you from the crowd when applying for jobs/interns.

# Say thanks

Thank the giants upon whose shoulders you stand.

# Volunteer for a talk

Effective public communication is critical to success.

# Practice writing

Effective writing is equally critical to success.

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- Keeping your committee informed eliminates surprises at your defense.

# Table of Contents

See an external link.

# Table of Contents

# Why do we need a good time mangement?

Be on top of things. Avoid drama and stress.

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Make EVERYTHING you plan to do as a calendar entry (including planning the calendar) and do it at that time.

# Calendar, not to-do lists: viewing time as space[2].

- Principle 1: Everything takes time. So everything needs to be on your calendar.
- Principle 2: It is easier to *measure how wrong your time estimates are* than *it is to fix them*.
- Principle 3: More generally, incorporate your patterns.
- Principle 4: Re-plan.
- Principle 5: Break it down.
- Principle 6: Backtrack. Foresee.
- Principle 7: Visualize your time.

[2] See details in https://deviparikh.medium.com/calendar-in-stead-of-to-do-lists-9ada86a512dd

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- If you are not a morning person
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- If you see repeatedly that you are too tired on Saturdays to do anything
  → Mark your entire Saturdays as “goof off”

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- The plan is feasible. But it is not guaranteed to work.
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- The future you will appreciate the extra buffer!

# Principle 5: Break it down.

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2. Break it down.
3. Mark them on your calendar in available slots

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- Re-planning needs empty slots, needs buffer.

# Principle 7: Visualize your time.

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$\downarrow$

We need • a productive routine and • a good time management.

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Doing regular exercise (e.g., in the early morning) gives you the

- mental acuity
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- emotional stability
- social health

# Eat that frog

Start your day by working on the most important task of the day (aka the frog)

# Stick to your plan and do what you are doing

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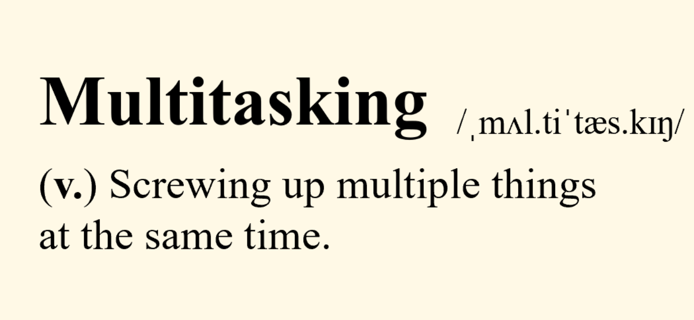
**Multitasking** /ˌmʌl.tiˈtæs.kɪŋ/

(v.) Screwing up multiple things at the same time.

Multi-tasking gives you the illusion of productivity.

# Stick to your plan and do what you are doing




Multi-tasking gives you the illusion of productivity.

Follow your calendar and focus on one thing at a time instead.

# Practice self-care

Take care of yourself!

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- Get enough sleep,

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Take care of yourself!

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# Practice self-care

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- Do exercise.

# Use interrupt coalescing

Group and defer interruptions

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Allocate a specific time slot in a day to address these interruptions.

# Learn to say no

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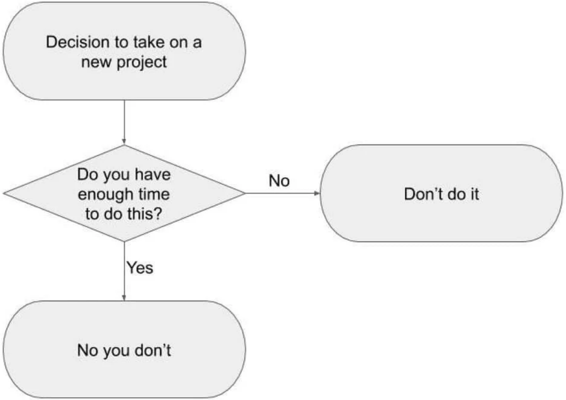

# Give up precise task prioritization

Sometimes you may be *spending more time ranking the tasks* than *actually doing it*.

# Touch each email/message exactly once

Open an email, skim through it, and make a quick decision
(reply/archive/allocate a time slot).

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- Get an ergonomic office chair.

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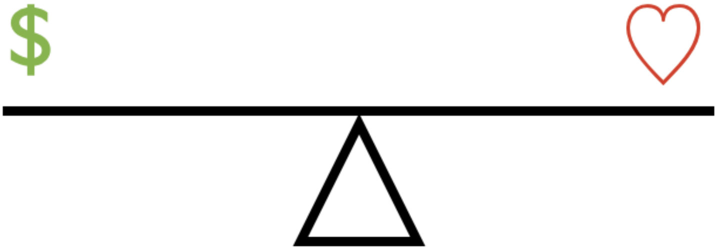
- Move your books into your work-space.
- Get an ergonomic office chair.
- Get a high-quality ergonomic keyboard.

# Table of Contents

# Tips for work-life balance



**Some high-level tips[3]:**

- Explicitly acknowledge priorities
- Avoid workaholism
- Avoid perfectionism
- Set and enforce boundaries
- Avoid over-commitment
- Use a work-flow system
- Communicate
- Keep hobbies
- Exploit opportunity cost
- Continuously adapt

[3] See details in https://matt.might.net/articles/work-life-balance/

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✗ If factors like “health”, “happiness”, or “family” rank below professional goals
$\Rightarrow$ your long-term priorities are not stable.

✗ If those inverted priorities persist
$\Rightarrow$ a crisis—injury/sickness, depression or divorce—is inevitable.

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✓ Efficiency.

The equation for work:

$$\text{output} = \underbrace{\text{unit of work / hour}}_{:=\text{productivity}} \times \text{hours worked} \tag{1}$$

# Avoid perfectionism

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Iterate toward perfection!

# Set and enforce boundaries

Maintaining work-life balance require setting boundaries.

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For example,

- I always schedule recurring “life” time into my calendar

# Avoid over-commitment

Learn when and how to say “no.”

# Keep hobbies

Even the best jobs create stress.

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Even the best jobs create stress.

*Hobbies* are a way of letting stress go before it explodes.

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Check out links below:

- How to Get a Great Letter of Recommendation
- Academic job search advice
- Fantastic Faculty Jobs and How to Get Them?
- See more at `https://github.com/jbhuang0604/awesome-tips#career`